Final exam of Biochemistry

Fill-in the blank questions

1. Compounds that contain a nitrogen base, a pentose sugar and a phosphate group are called ___. (answer: nucleotides)
2. Two purines found in DNA are ___. (answer: adenine, guanine)
3. Small, circular, extrachromosomal DNA molecules are ___. (answer: plasmids)
4. Restriction endonucleases that make staggered cuts produce ___. (answer: sticky ends)
5. The linking number for underwound DNA is ___ than that for relaxed DNA. (answer: less)
6. Regions of DNA that influence where gene transcription starts and ends and the rate of transcription; called ____ sequences. (answer: regulatory)
7. The replication ___ is a moving opening that leads the replication process. (answer: fork)
8. In bacteria, when DNA damage is extensive, the ___ response kicks in and initiates error-prone repair. (answer: SOS)
9. ___ supercoils form ahead of the transcription bubble. (answer: Positive)
10. Hairpin loops of nascent mRNA sometimes act as ___, abruptly terminating DNA transcription to regulate gene expression. (answer: attenuators)

Multiple choice questions

1. Which of the following is true for DNA? (answer is D)

a) Phosphate groups project toward the middle of the double helix.

b) Deoxyribose units are connected by 3',5'-phosphoanhydride bonds.

c) The 5' end of both strands are at the same end of the double helix.

d) G/C pairs share three hydrogen bonds.

2. In the DNA sequencing by the Sanger (dideoxy) method: (answer is D)

a) the template strand of DNA is radioactive.

b) enzymes are used to cut the DNA into small pieces, which are then separated by electrophoresis.

c) ddTTP is added to each of four reaction mixtures prior to synthesis of complementary strands.

d) the role of ddATP is to occasionally terminate synthesis of DNA where dT occurs in the template strands.

3. The phosphodiester bonds that link adjacent nucleotides in DNA: (answer is B)

a) are positively charged.

b) join the 3' hydroxyl of one nucleotide to the 5' phosphate of the next.

c) always link A with T and C with G.

d) are positively charged and always link A with T and C with G.

4. DNA supercoiling: (answer is A)

a) results in compaction of the DNA structure.

b) makes the DNA molecule relaxed.

c) occurs only in viruses.

d) usually results from overwinding.

5. Human chromosomes are extremely large, complex structures. Which of the following statements correctly describes the organization of human chromosomes? (answer is B)

a) All the genetic information of the cell is encoded in the nuclear, chromosomal DNA.

b) Genes for histone and ribosomal RNAs are structural genes.

c) Most of the chromosomal DNA codes for protein.

d) All the above statements are correct.

6. Consider the mRNA sequence: (5') AAUGCAGCUUUAGCA (3'). The sequence of the *coding* strand of DNA is: (answer is C)

a) (5') ACGATTTCGACGTAA (3').

b) (3') TTACGTCGAAATCGT (5').

c) (5') AATGCAGCTTTAGCA (3').

d) (3') AATGCAGCTTTAGCA (5').

7. What is found *within* the transcription bubble? (answer is D)

a) Positive supercoils in the DNA.

b) Negative supercoils in the DNA.

c) An intact DNA duplex.

d) A short DNA-RNA hybrid.

8. Prokaryotic *and* eukaryotic promoters: (answer is A)

a) contain AT-rich regions.

b) interact with transcription factors.

c) are influenced by base sequences thousands of base pairs away, which increase their activity.

d) have all of the above properties.

9. CRP has been found to contain a typical helix-turn-helix motif. Which of the following correctly describes the interactions between CRP and the *lac* operon. (answer is B)

a) Covalent bonds occur between the DNA helix and CRP helices.

b) Nonpolar interactions and hydrogen bonds contribute to the specificity of the CRP-DNA interactions.

c) The double helix of the operon DNA interacts with the double helix of the CRP DNA.

d) The base sequence of the DNA has no influence on the binding of CRP.

10. Which of the following statements about enhancer elements is correct. (answers are all)

a) They can be located on either side of the gene they activate.

b) They will not function unless located at some distance from the gene they control.

c) They bind transcription factors.

d) An enhancer element activating a specific gene is not functional in all cell types.

Cumulative

1. You wish to determine the relative degree of sequence similarity for DNA from different species of Galapagos finches (referred to as species A, B and C). You make hybrid duplexes of the DNA between A and B, A and C, and B and C. You measure the increase in the absorption of UV light with increasing temperature for each of the duplex solutions and obtain the following data:

Hybrid Duplex	t_m
A + B	80 °C
A + C	87 °C
B + C	83 °C

a) Which two species have a greater degree of similarity in their DNA sequences? Why?

Answer: The melting point of DNA provides information concerning the degree of hydrogen boning between two strands of DNA; the higher the melting point, the greater the extent of noncovalent bonding. Therefore, for the hybrid duplexes, a higher melting point indicates a greater

degree of sequence similarity between the two strands. The data suggest that species A and C share the highest degree of similarity in their DNA sequences.

b) What factors should be taken into account when interpreting these data?

Answer: Because G/C pairs share three hydrogen bonds, whereas A/T pairs share only two, DNA hybrids having a high proportion of G/C pairs will have a higher melting temperature than hybrids having more A/T pairs. Consequently, a hybrid with fewer matching bases that were all G/C pairs could theoretically have a higher melting temperature than a hybrid with more matches all occurring between A and T.

2. (a) The *Homo sapiens* genome has approximately 3×10^9 base pairs; if an "average" protein contains 400 amino acids, what is the maximum number of proteins that can be encoded by the human genome?

   Answer: If the average protein contains 400 amino acids, the minimum size for a gene is 1,200 base pairs. Dividing the total number of base pairs (3×10^9) by 1,200 gives a maximum of 2.5×10^6 proteins that can be encoded by the human genome.

   (b) Analysis of the sequence of the human genome suggests that there are 30,000-40,000 protein-coding genes, and it is estimated that human cells make ten times this number of proteins due to differential splicing, etc. Using the information from part (a), what percentage of the total genome is actually used to encode proteins? How would you explain the answers you obtained for (a) and (b)?

   Answer: In actuality, only about 1.6% of the genome is used to encode proteins. This number is obtained by determining the number of base pairs (bp) minimally needed to encode 40,000 protein-encoding genes (40,000 proteins × 1,200 bp per protein from part a above = 4.8×10^7) and then dividing that number by the total number of base pairs in the genome (3×10^9). The "extra" 98.4% is noncoding DNA containing a variety of types of sequences, including what some refer to as "junk" DNA. Also included in the extra DNA are regulatory regions (enhancers and promoters); spacer DNA; duplicated portions that are essentially redundant; pseudogenes; repetitious DNA; centromeric and telomeric regions; and other regions that remain to be identified.

   (c) The calculation made in part (b) above is an underestimation of the percentage of the genome used to encode proteins. Why?

Answer: The calculations made above assume that the genes are composed only of exons; introns were not included in our approximations of the amount of DNA needed to encode a protein. The average size of a human gene has been estimated to be much bigger than 1,200 bp (approximately 27,000 bp based on the transcribed region only). Using this number, the percent of the human genome used to encode proteins increases (to about 36%).